Estimation

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1) Population and Sample:

The group of objects under study for any statistical purpose is known as population.

A small part which represents the population is known as sample.

2) Parameter and statistics:

If the statistical measures such as mean, standard deviation, variance etc. are calculated from population, then they are called parameter where as if such statistical measures are calculated from sample then they are statistic.

Measure	Statistic (sample)	Parameter (population)
Size	n	N
Mean	$\overline{X}$	μ (Mu)
S.D	s	σ
Variance	s^2	σ^2
proportion	p	P
correlation coefficient	r	ρ (rho)

3) Point estimation and interval estimation:

If a single value used to estimate the true value of population parameter then it is known as point estimation.

Eg: Average age of Nepalese is 60 years

If the true value of population parameter is estimated by giving a certain interval with certain degree of confidance then it is known as interval estimation.

Eg: Average age of Nepalese i.e betn 55 years - 65 years

Confidence interval for estimating population mean

1) When population size 'N' is not given:

$$\text{Confidence interval} = \left[\bar{X} \pm Z_\alpha \, S.E(\bar{X})\right]$$
$$= \left[\bar{X} \pm Z_\alpha \frac{\sigma}{\sqrt{n}}\right]$$

2) When population size 'N' is given:

$$\text{Confidence interval} = \left[\bar{X} \pm Z_\alpha \, S.E(\bar{X})\right]$$
$$= \left[\bar{X} \pm Z_\alpha \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}\right]$$

where,

n = sample size

N = population size

$\bar{X}$ = Sample mean

σ = Population s.d

Z_α = value of Z_a + 1-α confidence level

$S.E(\bar{X})$ = standard error of mean

Note

1) If population s.d (σ) is not known the replace it b
sample s.d (s).

2) If confidence level isn't given then always take
$Z_\alpha = 3$.

Confidence Interval for estimating population ~~por~~ proportion :

1) When population size 'N' is not given :-

Confidence interval $= [p \pm Z_\alpha S.E(p)]$

$$= \left[p \pm Z_\alpha \sqrt{\frac{pq}{n}}\right]$$

2) When population size 'N' is given,

Confidence interval $= [p + Z_\alpha S.E(p)]$

$$= \left[p \pm Z_\alpha \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}\right]$$

where, p = sample population

$q = 1 - p$

P = population proportion

$Q = 1 - P$

Note

1) Character = Good / Bad
= Defective / not defective
= watch / not watch

2) जुन character को confidence interval निकाल्न खोजेको छ त्यही character को sample proportion 'p' हुन्छ।

Standard Error of Mean

Standard error of mean is a statistical term which used to study the deviation of sample mean from population mean.

Sample size for estimating population mean:

The sample size is

$$n = \left(\frac{Z_\alpha \, \sigma}{E}\right)^2$$

where, n = sample size
σ = population s.d
Z_α = value of Z at $1-\alpha$ confidence level
$E = |\bar{X} - \mu|$ = sampling error

Sample size for estimating population proportion:

The sample size is:

$$n = \left(\frac{Z_\alpha}{E}\right)^2 PQ$$

where, n = sample size
Z_α = value of Z at $1-\alpha$ confidence level
P = population proportion
$Q = 1-P$
$E = |P-p|$ = sampling error

* Types of tests under study

1) Z-test
2) t-test
3) P-test

} Parametic test

4) χ^2-test } Non-parametic

Population

Sample	Sample
$n > 30$	$n \leq 30$
Z-test	t-test

Z-test:

Z-test is a large sample test in which sample size is more than 30.

The main application of Z-test are:

1) Test of a single mean
2) Test of differences of two means
3) Test of a single proportion
4) Test of difference of two propotion.

Test of a single mean

To test $H_0 = \mu = \mu_0$ against suitable alternative hypothesis the test statistic is:

$$Z = \frac{\bar{x} - \mu}{S.E(\bar{x})}$$

$$\therefore Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where, $\bar{x}$ = sample mean
μ = population mean
σ = population s.d
n = sample size

Note: Its population s.d (σ) is not known then replace it by sample s.d (s).

Test of significance of a single proportion:

To test $H_0: P = P_0$ against suitable alternative hypothesis the test statistic is:

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$$Z = \frac{p - P}{S.E(p)}$$

$$\therefore Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

where, n = sample size

p = sample proportion

P = population proportion

$Q = 1 - P$

Note: कुनै character को बारेमा hypothesis set गरिएको छ

gurukul त्यही character को proportion sample proportion

Test of significance of double proportion:

To test $H_0: P_1 = P_2$ againsts suitable alternative hypothesi's, the test statistic is:

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where, n_1 = size of 1st sample
n_2 = size of 2nd sample
p_1 = sample proportion of 1st sample
p_2 = sample proportion of 2nd sample
$\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$
$\hat{Q} = 1 - \hat{P}$

Test of difference of two means:

$$H_0 = \mu_1 = \mu_2$$

$$Z = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Test of significance of two means:

To test $H_0: \mu_1 = \mu_2$ against suitable alternative hypothesi's, the test statistic is:

$$t = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t(n_1 + n_2 - 2)$$

where, n_1 = size of 1st sample
n_2 = size of 2nd sample
$\overline{X_1}$ = mean of 1st sample
$\overline{X_2}$ = mean of 2nd sample
S^2 = unbiased estimate of common population

Calculation of S^2

1) When sample s.d. S_1 and S_2 are given then,

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

2) when data given then,

$$S^2 = \frac{1}{n_1+n_2-2}\left\{ \sum X_1^2 - \left(\frac{(\sum X_1)^2}{n_1} + \sum X_2^2 - \frac{(\sum X_2)^2}{n_2} \right) \right\}$$

3) When sum of square of deviation from mean given th

$$S^2 = \frac{1}{n_1+n_2-2}\left\{ \sum (X_1 - \overline{X_1})^2 + \sum (X_2 - \overline{X_2})^2 \right\}$$

T-test

Paired t-test of difference of two means:

To test $H_0: \mu_1 = \mu_2$ against suitable alternative hypot the test statistic is:

$$t = \frac{\bar{d}}{\frac{S}{\sqrt{n}}} \sim t(n-1)$$

where, $\bar{d} = \frac{\Sigma d}{n}$

$d = X - Y$ = Difference of two sets of observation
n = no. of pair obs.

$$s = \sqrt{\frac{1}{n-1}\left\{\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right\}}$$

Note

In t-test of difference of two means, data are collected from two different sets of observations.

- In paired t-test of difference of two means data are collec twice from same set of observation.
- Paired t-test is used in the following cases:

* Test of productivity level of workers before and after traning program.
* Test of marks of students before and after tution class.
* Test of sales of product before and after advertisement campaign.
* Test of increase in reaction time before and after stimulus given.

F-test
(Variance Ratio Test)

F-test is a variance ratio test which is used to test whet two population variances are equal or not.

To test $H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$

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The test statistic is:

$$F = \frac{S_1^2}{S_2^2} \quad \text{if } S_2^2 > S_1^2$$

$$F = \frac{S_2^2}{S_1^2} \quad \text{if } S_2^2 > S_1^2$$

Calculation of S_1^2 and S_2^2

1) when data given:

$$S_1^2 = \frac{1}{n-1}\left\{\sum X_1^2 - \frac{(\sum X_1)^2}{n}\right\}$$

$$S_2^2 = \frac{1}{n-1}\left\{\sum X_2^2 - \frac{(\sum X_2)^2}{n}\right\}$$

2) When sum of squares of deviation from mean given

$$S_1^2 = \frac{1}{n-1} \sum (X_1 - \bar{X}_1)^2$$

$$S_2^2 = \frac{1}{n_2-1} \sum (X_2 - \bar{X}_2)^2$$

Note

F-test is used to test whether two population data are equally consistent / homogenous / stable / uniform or not.

$H_0: \sigma_1^2 = \sigma_2^2 \Rightarrow$ Data are similar

$H_1: \sigma_1^2 \neq \sigma_2^2 \Rightarrow$ Data are not similar

$F = \frac{S_1^2}{S_2^2} \sim F(n_1-1, n_2-1)$

$F = \frac{S_2^2}{S_1^2} \sim F(n_2-1, n_1-1)$

Analysis of Variance (ANOVA)

Analysis of variance is a statistical technique which is used to test the significant difference between three or more than 3 population means. There are two types of ANOVA:

1) One-way ANOVA:

If the influence of only one characteristic is studied at a time then it is known as one-way ANOVA.

2) Two-way ANOVA:

If the influence of two characteristics are studied at the same time, it is known as two ~~ya~~ way ANOVA.

<u>Note</u>

Total variation = Treatment + Error

Treatment ⇓ Main variable under study

Error ⇓ Unseen factors

Chi-square Test (χ^2-Test)

The main application of χ^2-test are:

1) ~~χ~~ χ^2 test of goodness of fit
2) χ^2-test of independence of attributes.

* χ^2-test of goodness of fit

This test is used to test the significant difference betwee observed frequency (experiment) and expected frequency (theory). This test is also used to test how well the observed data fits the theoritical distribution such as ~~bio~~ binomial distribution and poisson distribution.

To test H_0: There is no significant difference between observed frequency and expected frequency.
H_1: There is significant difference between observed frequency and expected frequency.

The test statistic is:

$$\chi^2 = \frac{\sum (O-E)^2}{E} \sim \chi^2_{(n-1)}$$

where, O = observed frequency
E = expected frequency

Calculation of Expected frequency:

1) $E = N \times$ Proportion (If ratio given)
2) $E = \frac{\sum O}{n}$ (If ratio not given)
3) $E = N \cdot {}^nC_r p^r \cdot q^{n-r}$ (For binomial distribution)
4) $E = \frac{N e^{-\lambda} \cdot \lambda^r}{r!}$ (For poisson distribution)

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Note

Degree of freedom is lost due to 3 reasons:

1) '1' d.f is lost due to $\Sigma O = \Sigma E (n-1)$

2) '1' d.f. is lost if we calculated value of 'p' as $np = \bar{X}$ for B.D and $\lambda = \bar{X}$ for P.D.

3) 1 or 2 or 3 d.f etc. is lost due to pooling. The expected frequency must be minimum 5. If expected frequency is less than 5 then we must pool it to make 5 or more than 5.

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χ^2-test of independence of attributes

This is used to test the relationship / association / variation between two variables. This test is used to test whether the values of one variable is dependent or independent on values of other variable.

To test Ho: There is no relationship between # and #

H_1: There is relationship between # and #

The test statistic is:

$$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{(r-1)(c-1)}$$

where, r = no. of rows

c = no. of columns

$$E = \frac{\text{Rows total} \times \text{column total}}{N}$$

Methods of data summarization — 10 marks

Correlation
Analysis of categorical data } — 10 marks (sure question)

Estimation (5 marks) sure

Hypothesis Testing
- Z-test
- t-test
- F-test

Non-parametric test
χ^2-test

10 marks